# MOSS CORE V1D — WORKING DESIGN

Integrated actuator cutoff, powered servo bus and speaker / external USB hub

PROTOTYPE — ERC/DRC do not validate analogue, firmware or real rover behavior.

01 USB Power

USB-C device. Current-limited input + reverse blocking + 3.3V buck. RF disabled. Encoder supply OFF before enumeration and in suspend. USB power limits require firmware and bench verification.

File: 01_USB_Power.kicad_sch

02 Controller

ESP32-S3-WROOM-1U-N8 only. Dedicated latch feedback, USB current fault and switched encoder power. USB ROM programming via BOOT/RESET; debug TX only. Firmware port required.

File: 02_Controller.kicad_sch

03 Hardware Interlock

V1D WORKING DESIGN. Corrected U10 physical pins. Hardware watchdog, NC loop and fault clear; added measured battery/rail supervision. Explicit recovery after faults. Not a certified emergency-stop system.

File: 03_Hardware_Interlock.kicad_sch

04 Drive Encoders

V1D WORKING DESIGN. All PWM and DIR outputs inhibit when actual motor power is invalid. U11 has four AND gates; U34 qualifies motor supply. Encoder power remains USB-budget controlled.

File: 04_Drive_Encoders.kicad_sch

05 Servo

V1D WORKING DESIGN. Native 3-pin powered servo connection. Half-duplex ioff buffers; switched idle bias and measured servo-supply permission. Current budget requires user confirmation and bench validation.

File: 05_Servo.kicad_sch

06 Audio

V1D WORKING DESIGN. Integrated MAX98357A, 9 dB gain, PH2 connector for user-confirmed 4-ohm/3-W speaker. Both output terminals are active. Firmware level limiting and load tests remain required.

File: 06_Audio.kicad_sch

07 Current Sense

3S pack only. External 2mOhm Kelvin shunt >=3W subject to derating. Fuse thin Kelvin wires AT SOURCE. INA226 measures total pack current, not individual motors.

File: 07_Current_Sense.kicad_sch

09 Audio Power

Dedicated nominal 5.077 V audio buck through F2 from BAT_PROTECTED; no positive connection to USB VBUS. MLCC DC-bias, fuse fault capacity, compensation, thermal and transient tests remain required.

File: 09_Audio_Power.kicad_sch

10 Battery Power

V1D WORKING DESIGN. Refer to the power and monitor contracts. Current limits, load capacitance, transient energy and board layout are not hardware validated.

File: 10_Battery_Power.kicad_sch

11 Motor Power

V1D WORKING DESIGN. Refer to the power and monitor contracts. Current limits, load capacitance, transient energy and board layout are not hardware validated.

File: 11_Motor_Power.kicad_sch

12 Servo Power

V1D WORKING DESIGN. Refer to the power and monitor contracts. Current limits, load capacitance, transient energy and board layout are not hardware validated.

File: 12_Servo_Power.kicad_sch

13 Rail Supervision

V1D WORKING DESIGN. Refer to the power and monitor contracts. Current limits, load capacitance, transient energy and board layout are not hardware validated.

File: 13_Rail_Supervision.kicad_sch

14 Power Fault Memory

V1D WORKING DESIGN. Refer to the power and monitor contracts. Current limits, load capacitance, transient energy and board layout are not hardware validated.

File: 14_Power_Fault_Memory.kicad_sch

15 Power Diagnostics

I2C 0x41 input diagnostics: actual motor/servo rail valid and latched rail losses. Configure all inputs, disable internal pull-ups, then verify registers before trusting readings. This device cannot authorize power.

File: 15_Power_Diagnostics.kicad_sch

Power flags identify the passive-fed rails only; they are not protection devices.




MOSS / Laurent & Codex

Sheet: /

File: MOSS_Core_V1D_DRAFT.kicad_sch

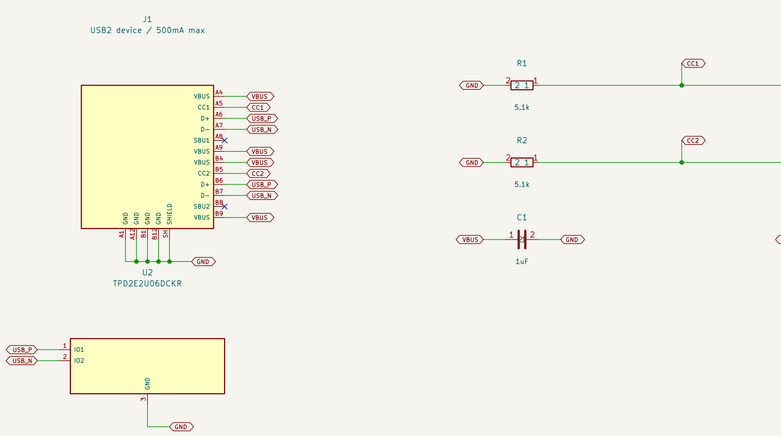
**Title: MOSS Core V1D DRAFT / system overview**

| Size: A2 | Date: 2026-10-15 | **Rev: V1D DRAFT** |
|---|---|---|
| KiCad E.D.A. 10.0.6 | | Id: 1/15 |

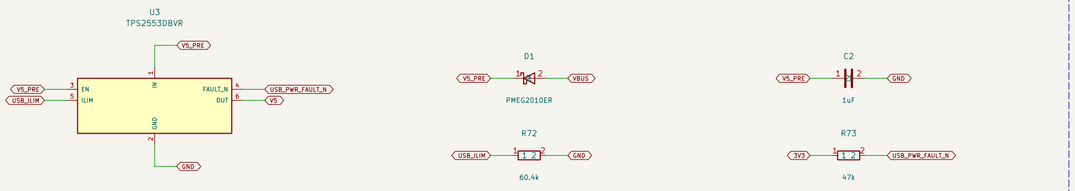
# 01 USB Power

USB-C device. Current-limited input + reverse blocking + 3.3V buck. RF disabled. Encoder supply OFF before enumeration and in suspend. USB power limits require firmware and bench verification.

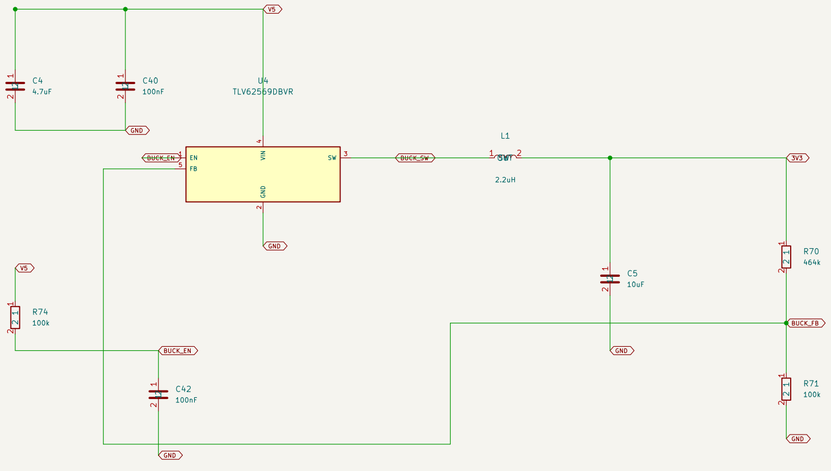
## USB-C, CC and ESD




## Host reverse blocking and current limit



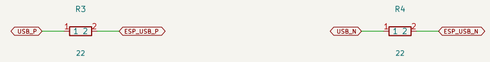

## 3.3 V buck / enable / feedback




3.384 V nominal: 0.6 V × (1 + 464k / 100k).

Keep SW loop small; feedback away from SW. Local GND vias.

## USB series resistors, at MCU

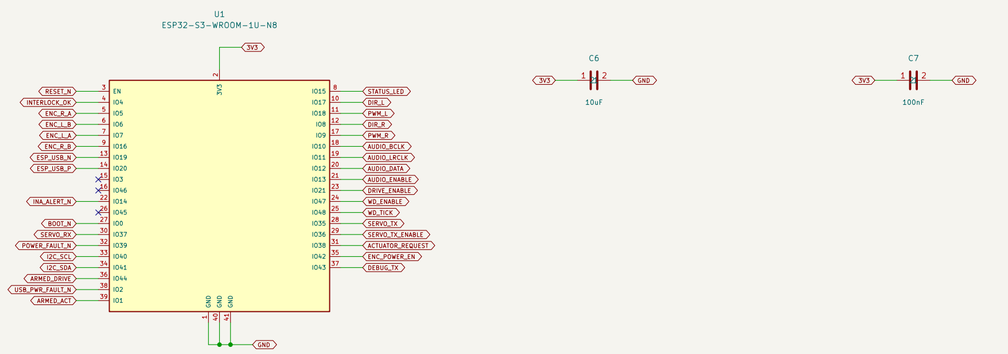



| | |
|---|---|
| MOSS / Laurent & Codex | |
| Sheet: /01 USB Power/ | |
| File: 01_USB_Power.kicad_sch | |
| **Title: MOSS Core V1D DRAFT / 01 USB Power** | |
| Size: A2 | Date: 2026-10-15 |
| KiCad E.D.A. 10.0.6 | **Rev: V1D DRAFT** |
| | Id: 2/15 |

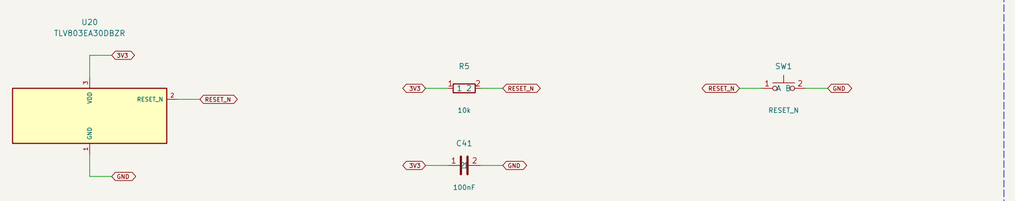
# 02 Controller

ESP32-S3-WROOM-1U-N8 only. Dedicated latch feedback, USB current fault and switched encoder power, USB ROM programming via BOOT/RESET; debug TX only. Firmware port required.

## ESP32-S3 module / local supply



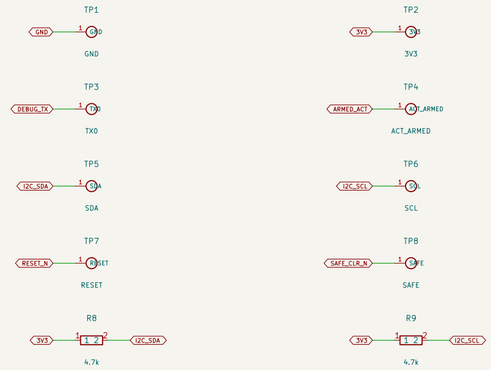

## Supervised reset



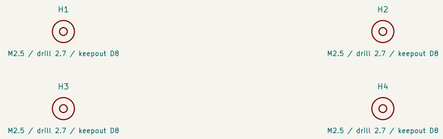

## BOOT and status LED




## I2C pull-ups and service pads




## Mechanical interface: M2.5, D8 reservation

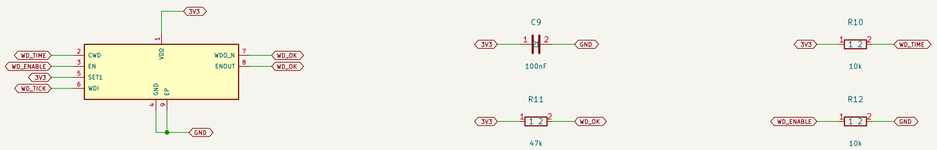



MOSS / Laurent & Codex
Sheet: /02 Controller/
File: 02_Controller.kicad_sch
Title: MOSS Core V1D DRAFT / 02 Controller
Size: A2 | Date: 2026-10-15 | Rev: V1D DRAFT
KiCad E.D.A. 10.0.6 | Id: 3/15

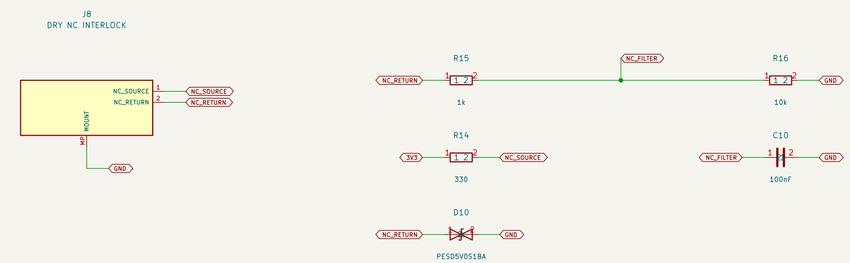
# 03 Hardware Interlock

V1D WORKING DESIGN. Corrected U10 physical pins. Hardware watchdog, NC loop and fault clear; added measured battery/rail supervision. Explicit recovery after faults. Not a certified emergency-stop system.

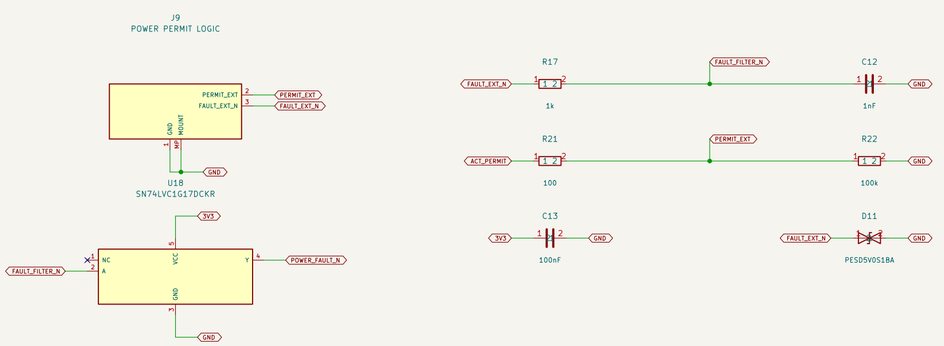
## External watchdog / 200 ms nominal



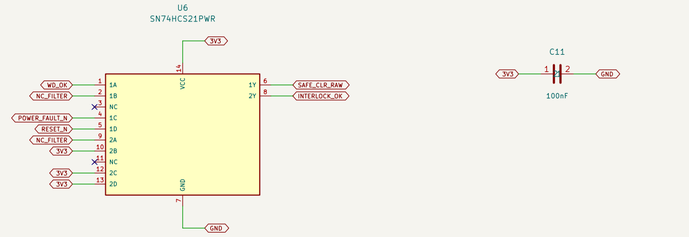

## Normally closed contact input



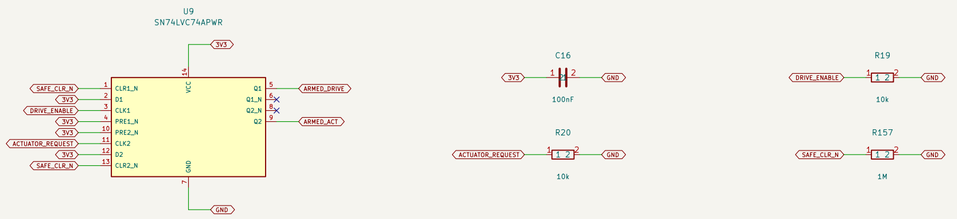

## External power permission and fault receiver



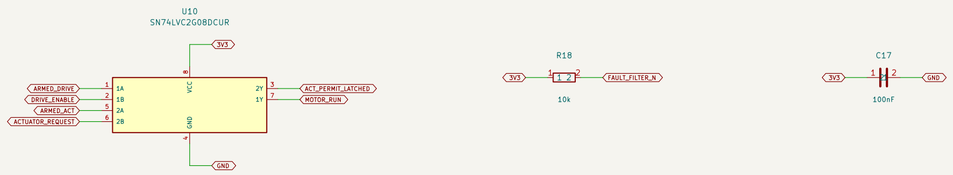

## Healthy gate: WD / NC / fault / reset




## Fresh-edge arming latches




## Requested and latched permissions




MOSS / Laurent & Codex

Sheet: /03 Hardware Interlock/

File: 03_Hardware_Interlock.kicad_sch

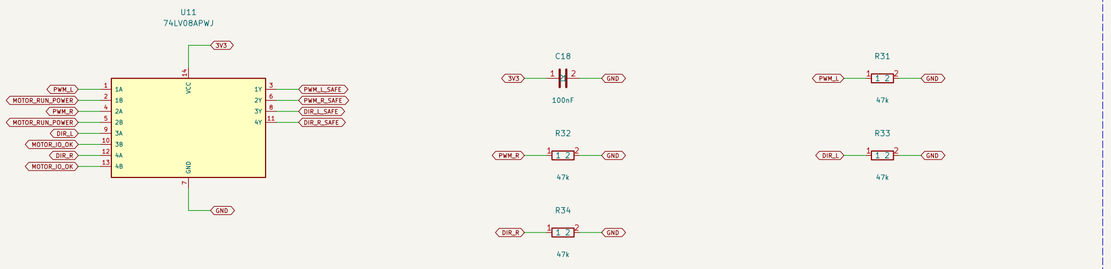
**Title: MOSS Core V1D DRAFT / 03 Hardware Interlock**

| Size: A2 | Date: 2026-10-15 | **Rev: V1D DRAFT** |
|---|---|---|
| KiCad E.D.A. 10.0.6 | | Id: 4/15 |

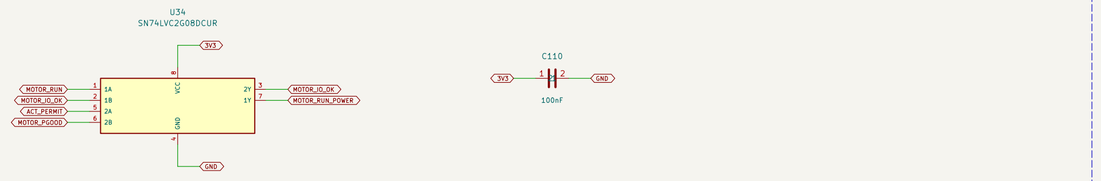
# 04 Drive Encoders

V1D WORKING DESIGN. All PWM and DIR outputs inhibit when actual motor power is invalid. U11 has four AND gates; U34 qualifies motor supply. Encoder power remains USB-budget controlled.

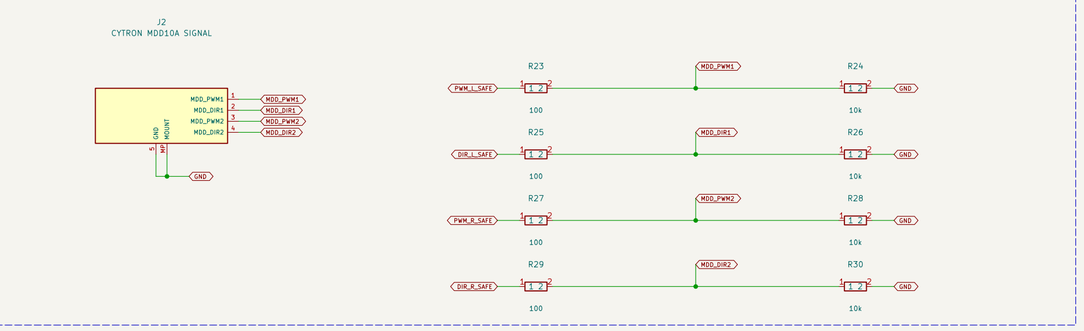
## Motor input defaults and four-channel mask



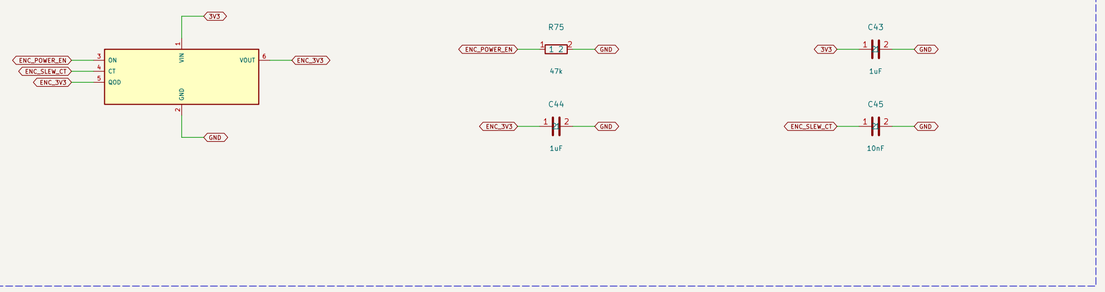

## Actual motor rail qualification



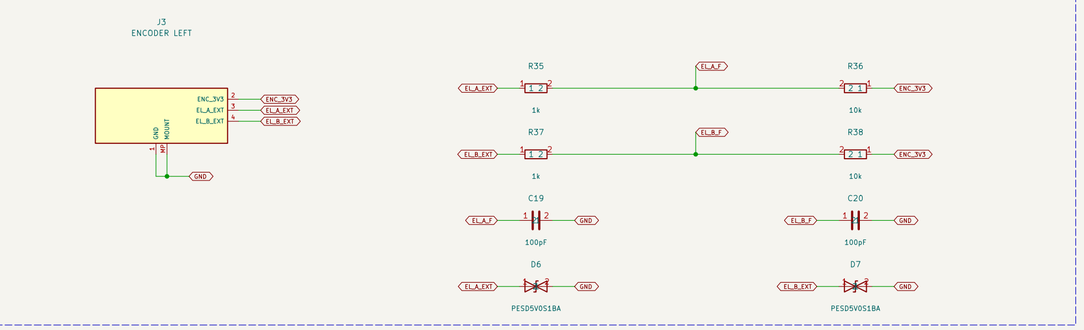

## Cytron PWM and DIR logic connector



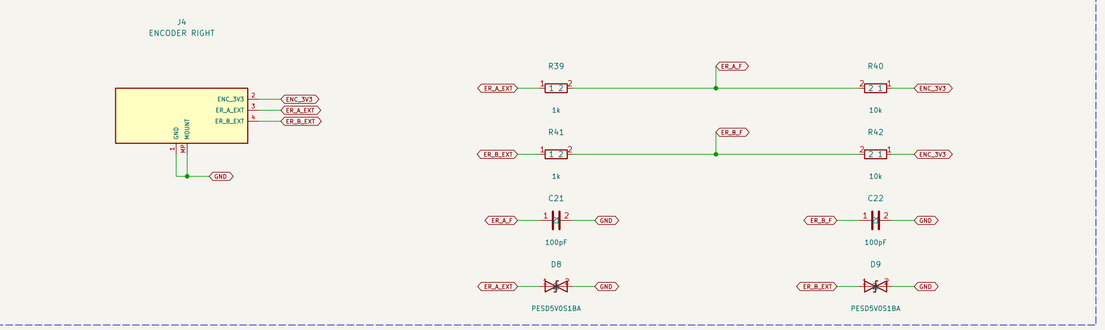

## Encoder supply OFF until USB configured



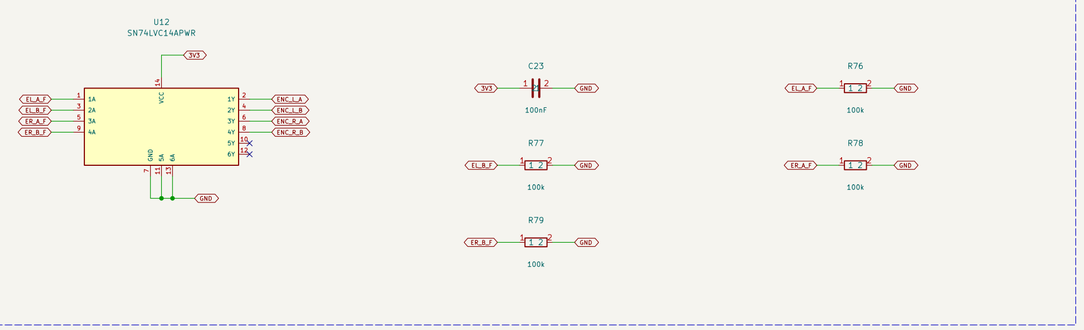

## Left encoder / filters / ESD




## Right encoder / filters / ESD




## Encoder Schmitt receivers




MOSS / Laurent & Codex
Sheet: /04 Drive Encoders/
File: 04_Drive_Encoders.kicad_sch

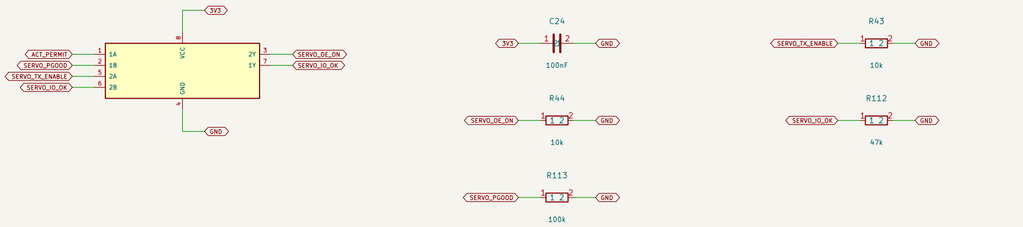
**Title: MOSS Core V1D DRAFT / 04 Drive Encoders**

| Size: A2 | Date: 2026-10-15 | **Rev: V1D DRAFT** |
|---|---|---|
| KiCad E.D.A. 10.0.6 | | Id: 5/15 |

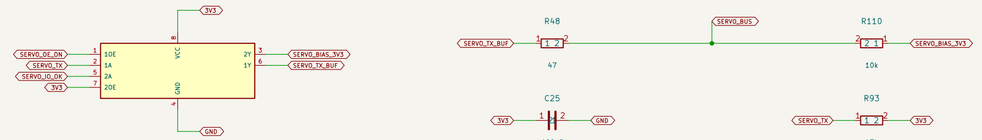
# 05 Servo

V1D WORKING DESIGN. Native 3-pin powered servo connection. Half-duplex ioff buffers; switched idle bias and measured servo-supply permission. Current budget requires user confirmation and bench validation.

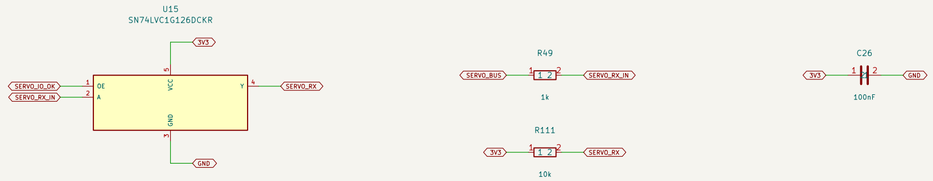
## Measured supply and TX permission




## Half-duplex transmitter and switched idle bias




## Receiver and idle level on MCU side




## Native powered servo output




**MOSS / Laurent & Codex**

Sheet: /05 Servo/
File: 05_Servo.kicad_sch

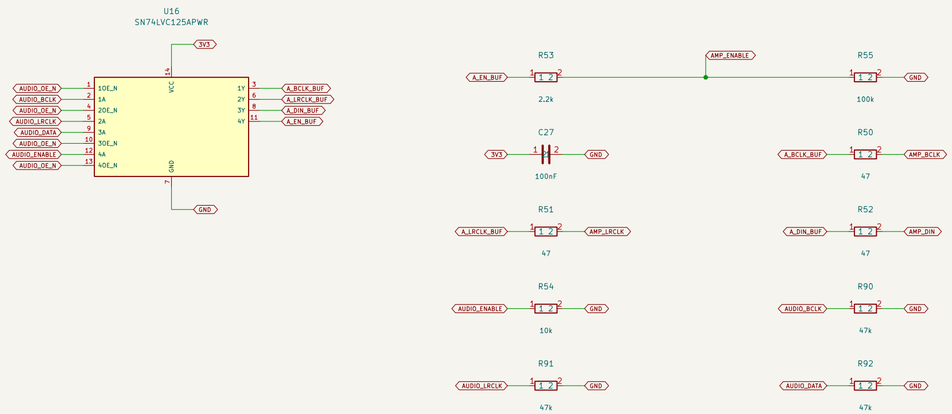
***Title: MOSS Core V1D DRAFT / 05 Servo***

| Size: A2 | Date: 2026-10-15 | **Rev: V1D DRAFT** |
|---|---|---|
| KiCad E.D.A. 10.0.6 | | Id: 6/15 |

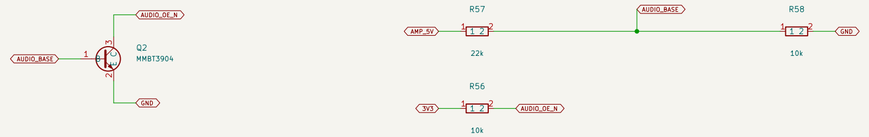
# 06 Audio

V1D WORKING DESIGN. Integrated MAX98357A, 9 dB gain, PH2 connector for user-confirmed 4-ohm/3-W speaker. Both output terminals are active. Firmware level limiting and load tests remain required.

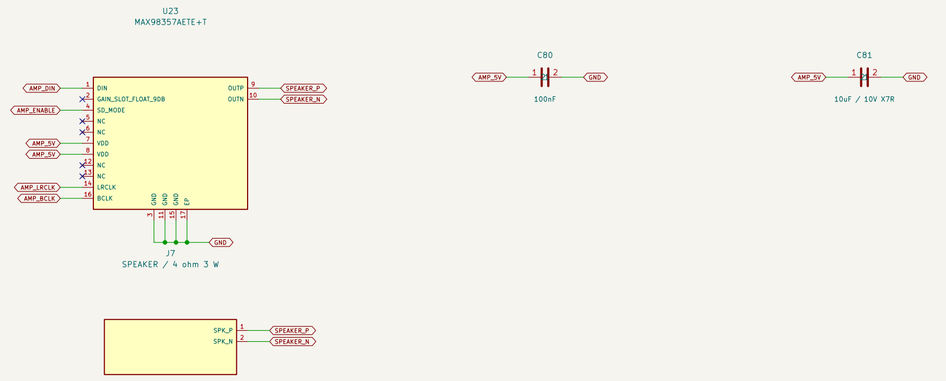
## I2S isolation and idle states




## On-board amplifier supply detection




## Integrated speaker amplifier







**MOSS / Laurent & Codex**

Sheet: /06 Audio/
File: 06_Audio.kicad_sch

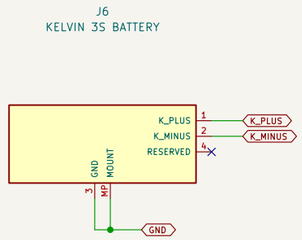
***Title: MOSS Core V1D DRAFT / 06 Audio***

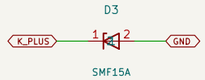
| Size: A2 | Date: 2026-10-15 | **Rev: V1D DRAFT** |
|---|---|---|
| KiCad E.D.A. 10.0.6 | | Id: 7/15 |

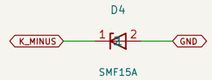
# 07 Current Sense

3S pack only. External 2mOhm Kelvin shunt >=3W subject to derating. Fuse thin Kelvin wires AT SOURCE. INA226 measures total pack current, not individual motors.

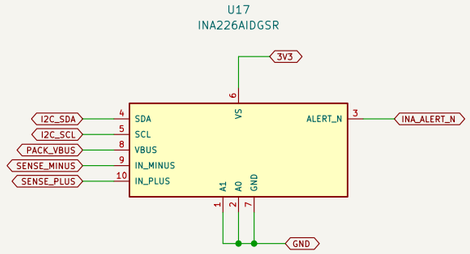
## Kelvin entry / source protection external

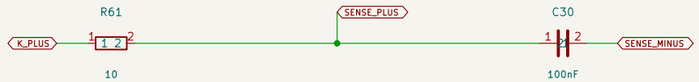



## INA226 / input filters / I2C



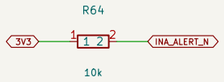













**MOSS / Laurent & Codex**

Sheet: /07 Current Sense/
File: 07_Current_Sense.kicad_sch

***Title: MOSS Core V1D DRAFT / 07 Current Sense***

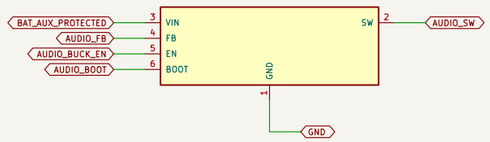
| | | |
|---|---|---|
| Size: A3 | Date: 2026-10-15 | **Rev: V1D DRAFT** |
| KiCad E.D.A. 10.0.6 | | Id: 8/15 |

# 09 Audio Power

Dedicated nominal 5.077 V audio buck through F2 from BAT_PROTECTED; no positive connection to USB VBUS. MLCC DC-bias, fuse fault capacity, compensation, thermal and transient tests remain required.

## Dedicated audio supply

U24
TPS54302DDCR




L2
10uH / 7.6A Isat typ

C84
100nF

R102
100k / 0.1%

R103
13.3k / 0.1%

R104
511k

R105
105k

C82
10uF / 50V X7R

C83
100nF / 50V X7R

C85
22uF / 10V X7R

C86
22uF / 10V X7R

C87
75pF / C0G

F2
1A / 125V fast

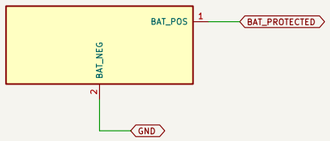
| | |
|---|---|
| MOSS / Laurent & Codex | |
| Sheet: /09 Audio Power/ | |
| File: 09_Audio_Power.kicad_sch | |
| ***Title: MOSS Core V1D DRAFT / 09 Audio Power*** | |
| Size: A3 | Date: 2026-10-15 |
| KiCad E.D.A. 10.0.6 | Id: 9/15 |

# 10 Battery Power

V1D WORKING DESIGN. Refer to the power and monitor contracts. Current limits, load capacitance, transient energy and board layout are not hardware validated.

## J10 / FUSED BAT INPUT /3S

J10
FUSED BAT INPUT /3S




D32




SMBJ13A-E3/52

MOSS / Laurent & Codex

Sheet: /10 Battery Power/
File: 10_Battery_Power.kicad_sch

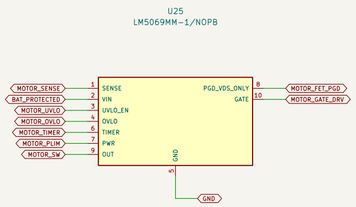
**Title: MOSS Core V1D DRAFT / 10 Battery Power**

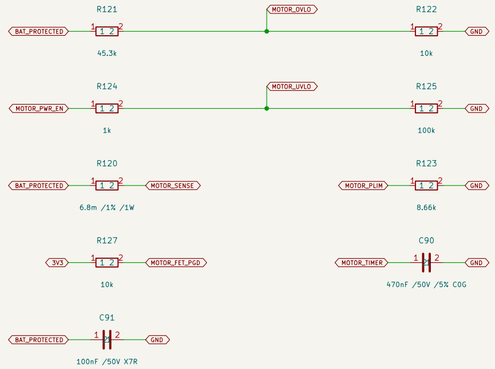
| Size: A3 | Date: 2026-10-15 | **Rev: V1D DRAFT** |
|---|---|---|
| KiCad E.D.A. 10.0.6 | | Id: 10/15 |

# 11 Motor Power

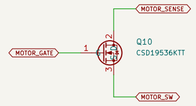
V1D WORKING DESIGN. Refer to the power and monitor contracts. Current limits, load capacitance, transient energy and board layout are not hardware validated.

## U25 / LM5069MM-1/NOPB



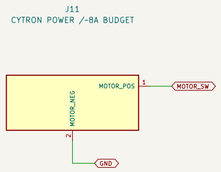

## Q10 / CSD19536KTT







## J11 / CYTRON POWER /-8A BUDGET




MOSS / Laurent & Codex

Sheet: /11 Motor Power/
File: 11_Motor_Power.kicad_sch

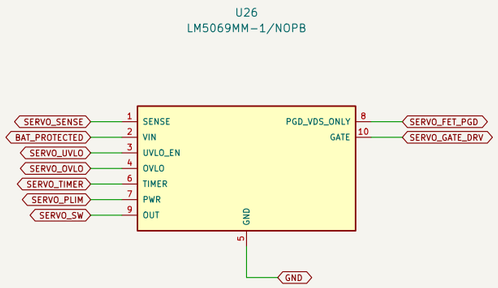
**Title: MOSS Core V1D DRAFT / 11 Motor Power**

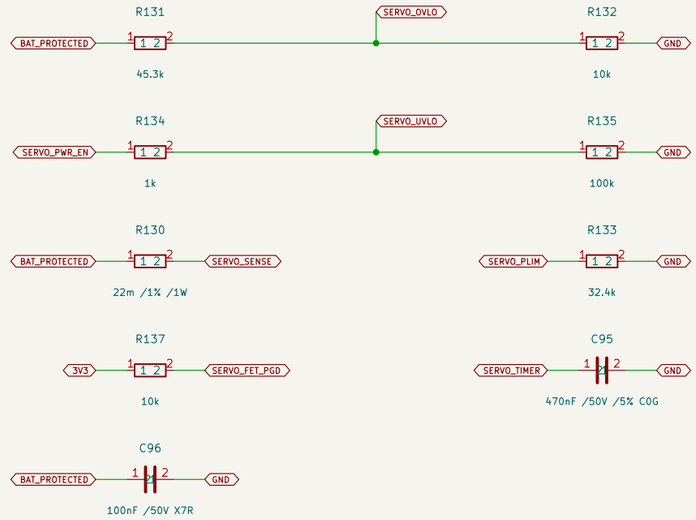
| Size: A2 | Date: 2026-10-15 | Rev: V1D DRAFT |
|---|---|---|
| KiCad E.D.A. 10.0.6 | | Id: 11/15 |

# 12 Servo Power

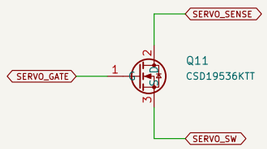
V1D WORKING DESIGN. Refer to the power and monitor contracts. Current limits, load capacitance, transient energy and board layout are not hardware validated.

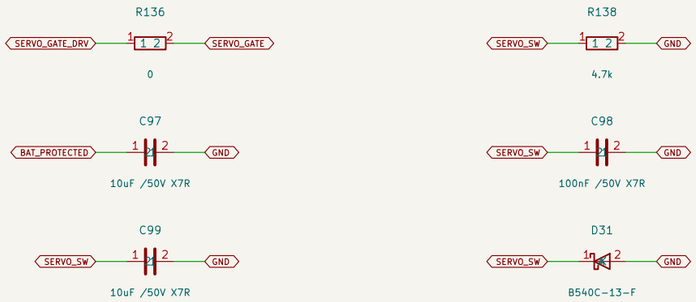
## U26 / LM5069MM-1/NOPB




## Q11 / CSD19536KTT




MOSS / Laurent & Codex

Sheet: /12 Servo Power/
File: 12_Servo_Power.kicad_sch

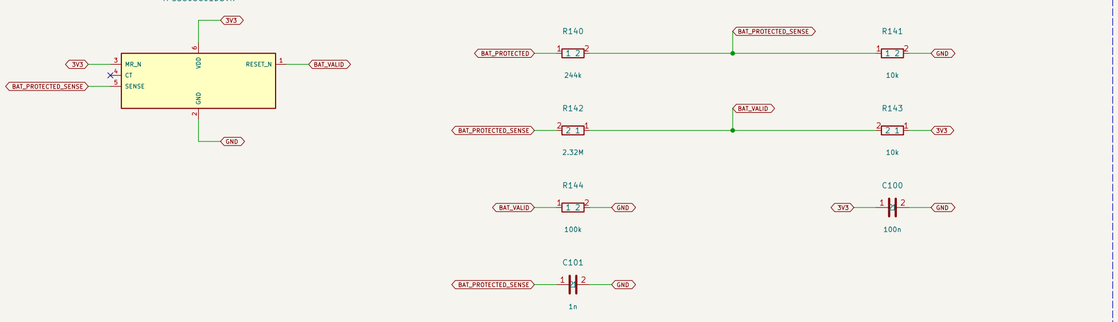
**Title: MOSS Core V1D DRAFT / 12 Servo Power**

| Size: A3 | Date: 2026-10-15 | Rev: V1D DRAFT |
|---|---|---|
| KiCad E.D.A. 10.0.6 | | Id: 12/15 |

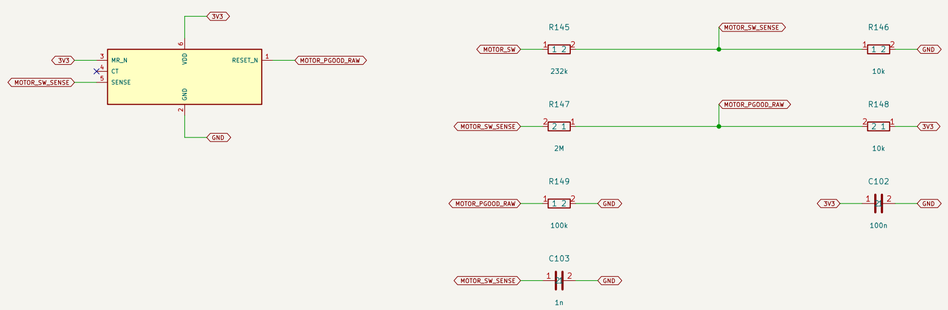
# 13 Rail Supervision

V1D WORKING DESIGN. Refer to the power and monitor contracts. Current limits, load capacitance, transient energy and board layout are not hardware validated.

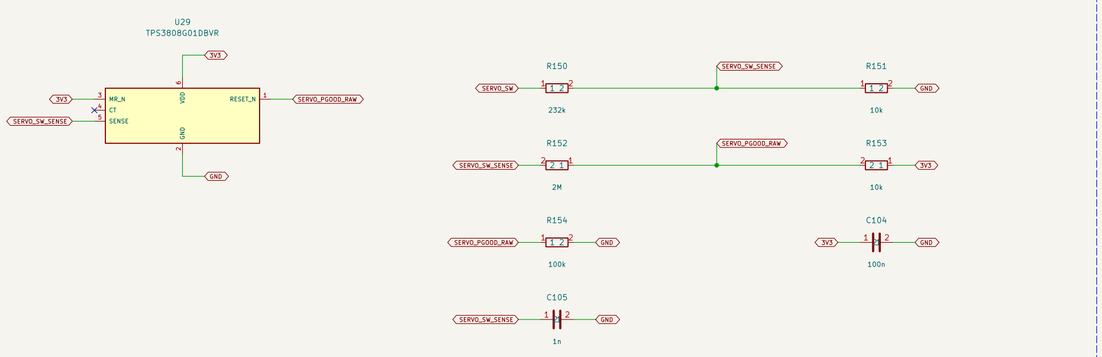
## Battery voltage valid




## Motor output voltage valid




## Servo output voltage valid



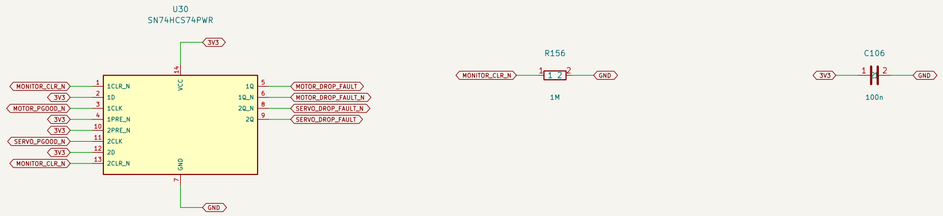

| | |
|---|---|
| MOSS / Laurent & Codex | |
| Sheet: /13 Rail Supervision/ | |
| File: 13_Rail_Supervision.kicad_sch | |
| **Title: MOSS Core V1D DRAFT / 13 Rail Supervision** | |
| Size: A2 | Date: 2026-10-15 |
| KiCad E.D.A. 10.0.6 | Id: 13/15 |

**Rev: V1D DRAFT**

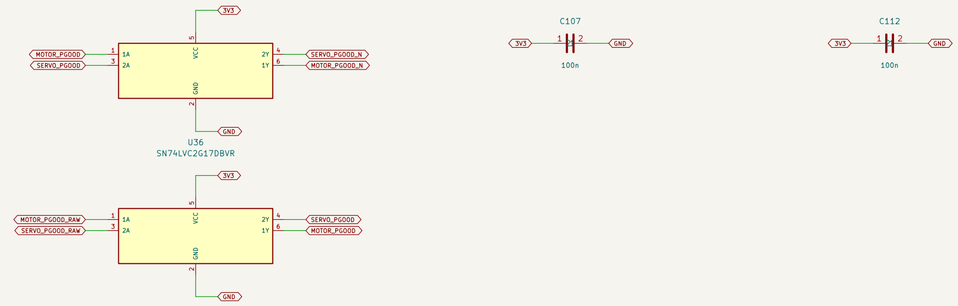
# 14 Power Fault Memory

V1D WORKING DESIGN. Refer to the power and monitor contracts. Current limits, load capacitance, transient energy and board layout are not hardware validated.

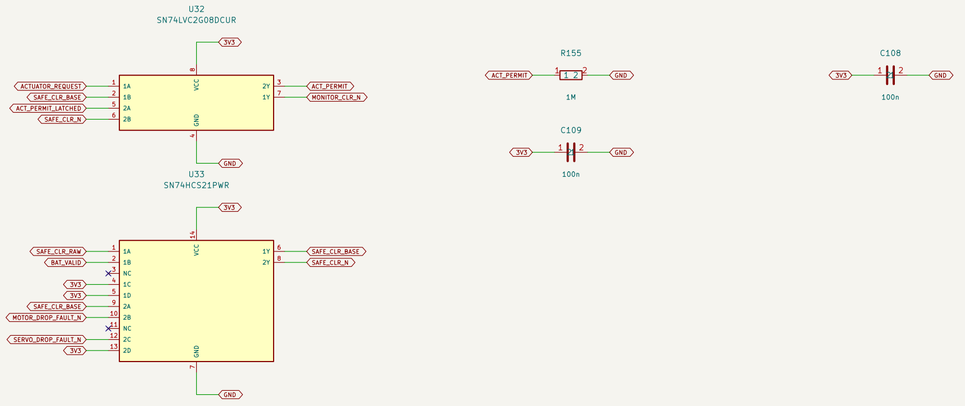
## Sticky rail-loss memory



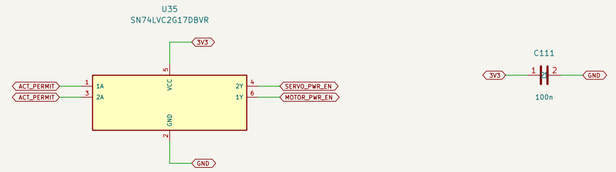

## Schmitt-conditioned rail events




## Direct hardware health and request gating




## Independent power-enable output buffers




MOSS / Laurent & Codex

Sheet: /14 Power Fault Memory/
File: 14_Power_Fault_Memory.kicad_sch

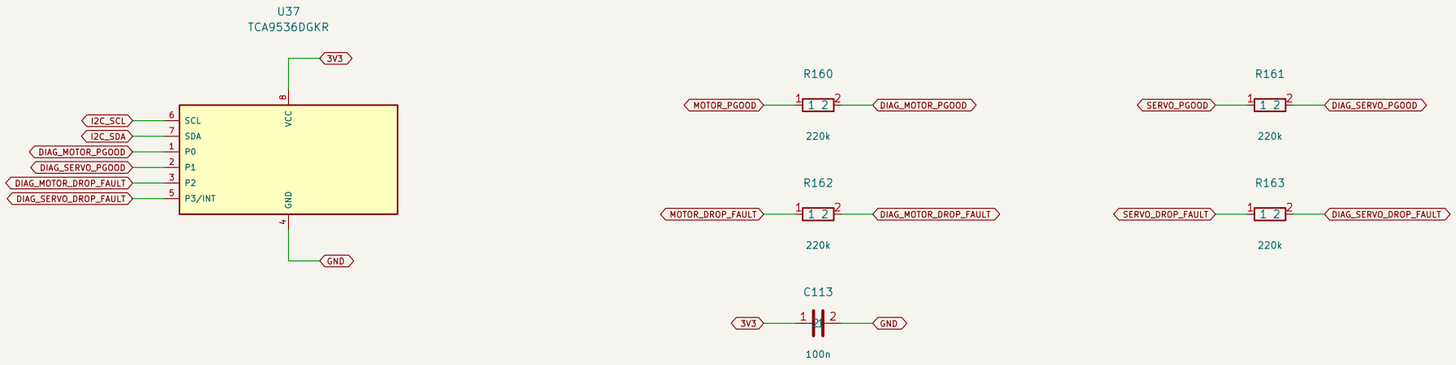
**Title: MOSS Core V1D DRAFT / 14 Power Fault Memory**

| Size: A2 | Date: 2026-10-15 | **Rev: V1D DRAFT** |
|---|---|---|
| KiCad E.D.A. 10.0.6 | | Id: 14/15 |

# 15 Power Diagnostics

I2C 0x41 input diagnostics: actual motor/servo rail valid and latched rail losses. Configure all inputs, disable internal pull-ups, then verify registers before trusting readings.
This device cannot authorize power.

## U37 / TCA9536DGKR




MOSS / Laurent & Codex

Sheet: /15 Power Diagnostics/
File: 15_Power_Diagnostics.kicad_sch

***Title: MOSS Core V1D DRAFT / 15 Power Diagnostics***

| Size: A3 | Date: 2026-10-15 | **Rev: V1D DRAFT** |
|---|---|---|
| KiCad E.D.A. 10.0.6 | | Id: 15/15 |